Flood situation report

Area: Teesta glacial lake flood, Sikkim, 4 Oct 2023 | Event date: 4 October 2023 | Latest radar image: 14 October 2023

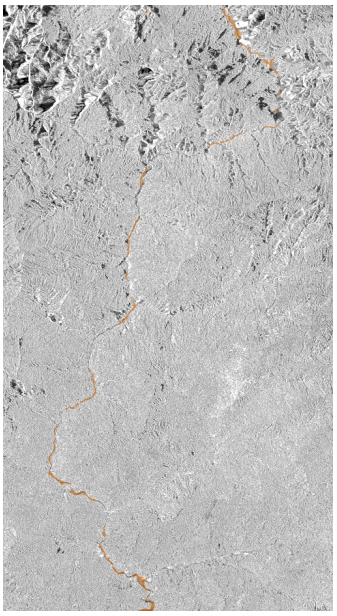
1. Where it flooded

The radar shows 859 ha (8.59 km²) flooded or buried by debris. F010 F011

Of this, 147 ha is open water and 712 ha is debris or scour. F012 F014

The radar could not see 4,915 ha because of terrain shadow, so conditions there are unknown. F016

The places with the most likely destroyed buildings: Pegong, Sangkalang, Dikchu, Majitar, Simchuthang. F057



Radar change map, before and after the event.

2. What was hit

Buildings: 490 likely destroyed and 77 possibly damaged, out of 8,835 mapped. F025 F021 F022

Roads: 79.1 km likely destroyed and 20.3 km possibly damaged, out of 2,268.0 km mapped. F038 F034 F035

Bridges: 18 likely destroyed (North Sikkim Highway, Sirwani Bridge) and 6 possibly damaged, out of 139 mapped. F031 F027 F028 F033

3. Who is cut off

9 of 41 settlements are likely cut off by road. F059 F058

Among them: Pegong, Dikchu, Lingthem, Passingdong, Bringbong. F063

The nearest health facility for 9 of the cut-off settlements is Mangan District Hospital. F064 F065

OpenStreetMap lists 99 health facilities in the area. Hospitals among them: 9. F323 F324

Repair priority

1. Reopening unnamed secondary bridge (bridge) reconnects 7 settlements. F326 F327 F330
2. Reopening unnamed trunk (road) reconnects 5 settlements. F332 F333 F335
3. Reopening unnamed unclassified (road) reconnects 1 settlement. F337 F338 F340
4. Reopening North Sikkim Highway bridge (bridge) reconnects 1 settlement. F342 F343 F346

Method

Sentinel-1 radar images from before and after the event are compared to map new water and debris. Pre-event OpenStreetMap roads, bridges and buildings are overlaid, and a road network model finds settlements that can no longer reach a health facility.

Educational prototype, not an operational tool. Numbers come from satellite analysis and have not been checked on the ground.

Grey codes such as F012 point to entries in facts.json.

Contains modified Copernicus Sentinel data 2026.
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